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A Novel Fully Quaternion based Nonlinear Attitude and Position Controller

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In this paper, a novel fully quaternion based second order stable error dynamics controller is developed to address the attitude tracking problem in general and the position tracking problem of a fixed pitch quadrotor in particular. Since the singularity associated with euler angle representation is avoided using a unit quaternion approach, our algorithm finds its advantage in controlling highly maneuverable systems like quadrotors. Our controller is also well suited for rigid systems like satellites which require tracking of large-angle amplitude trajectories. Numerical simulations show good tracking performance using the proposed control methods for satellites and quadrotors. Experimental results on commercially available quadrotor using open-source architecture validate the novel attitude controller.

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I. Introduction

Designing smooth control inputs for rigid body attitude and translation dynamics has long been a problem in focus since past few decades. Various approaches proposed in the past to fulfilling this objective can primarily be divided into two domains: one focusing on developing suitable control inputs, with PID being the most applied one, and the other focused on choosing better representation of state variables. Much of current research can be found combining these new approaches in both or either of these domains and studying its efficacy. The synthesis of control inputs involve linear control methods such as PID and its variants and nonlinear control methods such as Feedback Linearization, Nonlinear Dynamic inversion, and Sliding Mode and other adpative control theories. The translation dynamics representation has been done mostly in Euclidean space for Earth bound systems and in Polar or spherical coordinates for outer space systems. As for the orientation representation, the attitude of a rigid body belongs to the configuration space known as Special Orthogonal group SO(3) and is represented in most general terms as 3×3 rotation matrix.[1] The space can, however, be parameterized in terms of fewer parameters. The most used representation is the Euler angle formulation. However, it suffers from gimbal lock which results in a singularity in the formulation. Recently quaternions and rotation matrix have become the preferred tool to overcome the singularity of Euler angles. The applications of these approaches have been validated on various systems such as satellites, aircrafts and more popularly on UAVs which, due to their small size, also provide the advantage of providing cheap experimental platform for concept verification

Among these small Unmanned Aerial Vehicles, Quadrotors have recently gained attention for their usefulness in multi-agent missions, mapping and localization, acrobatic performances, and as autonomous payload delivery systems. They are mechanically simple however, their inherent instability and nonlinear dynamics make their control challenging. Various control strategies have been designed which enable systems to perform aggressive maneuvers such as multiple flips,[2] flight through narrow openings,[3] etc. Most of these approaches use hierarchical structures and divide the problem into two loops of position and attitude control.

Emil and George[4] proposed a nonlinear P^2 controller with the commanded moment proportional to the vector part of the quaternion error and actual angular velocity. Tayebi and McGilvray[5] designed a similar

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PD^2 feedback structure for the hover equilibrium point stabilization based on quaternion, quaternion velocity and angular velocity. The Lyapunov function used for proving stability was obtained in the same manner as the backstepping approach. A quaternion based nonlinear robust feedback controller was proposed by Xian et al[6] in which they stabilize the second order error dynamics in attitude and altitude while using filters to estimate the angular velocities. Their controller however requires inversion of a Jacobian matrix which becomes singular if first component of error quaternion is zero. A similar controller was also designed by Bangura and Mahony[7] with moment as a function of quaternion and angular velocity error.

In this paper we impose second order stable dynamics for the error quaternion in designing the attitude controller. Since we control second derivative instead of first, the control inputs are expected to be smoother than first order error dynamics which directly controls the first derivative or the direction of response.To the best of our knowledge most of the previous quaternion feedback controllers including those mentioned above are dependent only on rate of vector part of error quaternion vector for the feedback. However, it is our understanding that the scalar part can also also guide how fast the error dynamics converges. We also observed experimentally that during the gain tuning process the scalar part contribution can affect smoothness of the controller. Also, the derivative of quaternion has a nonlinear relationship with the angular velocity and hence our feedback, which is function of quaternion derivative rather than the angular velocity directly, is also unlike most of the previously proposed controllers which have angular velocity feedback.

Our attitude controller holds in general for any rigid body and can be applied to many significant areas. For example, with the growth in number and operational requirements of rigid body spacecraft, systems often have a need for highly accurate pointing maneuvers in numerous tasks of practical importance like weather monitoring, communications, survey of resources etc in which there arises a need to maintain a satellite in a fixed orientation with respect to the Earth. B. T. Costic et al[8] have addressed the attitude tracking problem for a rigid spacecraft using unit quaternion based adaptive approach that obtains asymptotic attitude tracking without the need for angular velocity measurements. In this paper we present simulation results by implementing the proposed second order error dynamics quaternion based attitude controller for achieving the desired satellite orientation.

The rest of the paper is organized as follows: A brief introduction to quaternion formulations is presented in Section II. Section III presents the dynamic models of various rigid bodies in for whom the controller has been designed. Section IV describes the attitude controller, which in general holds for any rigid body, and Section V describes the complete trajectory tracking algorithm for the quadrotor. Simulation results are then presented in Section VI and experimental results for the attitude controller are presented in Section VII. Finally we state some important conclusions and future work in Section VIII.

II. Quaternion Preliminaries

Quaternions[9–11] are hyper complex number of rank 4 which consists of a scalar part q_0 and vector part $\vec{q}$

$$\mathbf{q} = \begin{bmatrix} q_0 \\ \vec{q} \end{bmatrix} = [q_0 \ q_1 \ q_2 \ q_3]^T \tag{1}$$

When using them to represent the attitude of a body, they have the property of unity norm, i.e., $|\mathbf{q}| = 1$. The quaternion $\mathbf{q}$ in this paper will be reserved for representing attitude except when stating identities which may hold for more general cases. We will be following the Hamilton convention of quaternion, according to which the transformation between body and world frame is given by

$$\mathbf{x}^i = \mathbf{q} \circ \mathbf{x}^b \circ \mathbf{q}^* \tag{2}$$

where $\mathbf{x} = \left[0 \ \vec{x}^T\right]^T$ is the quaternion corresponding to any vector $\vec{x}$. Throughout the paper the bold counterpart of any vector will represent its corresponding quaternion. The quaternion product operator $\circ$ is defined as

$$\mathbf{q} \circ \mathbf{p} = \begin{bmatrix} p_0 q_0 - \vec{p} \cdot \vec{q} \\ q_0 \vec{p} + p_0 \vec{q} + \vec{q} \times \vec{p} \end{bmatrix} \tag{3}$$

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The rotation matrix $\mathbf{R}_{\mathcal{B}}^{\mathcal{E}} = (R(\mathbf{q}))$, transforming a vector from body from local body frame $\mathcal{B}$ to global Earth frame $\mathcal{E}$ is then given by

$$R(\mathbf{q}) = \begin{bmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 - q_0q_3) & 2(q_0q_2 + q_1q_3) \\ 2(q_1q_2 + q_0q_3) & (q_0^2 - q_1^2 + q_2^2 - q_3^2) & 2(q_2q_3 - q_0q_1) \\ 2(q_1q_3 - q_0q_2) & 2(q_0q_1 + q_2q_3) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{bmatrix} \tag{4}$$

The rate of change of quaternion $\dot{q}$ can be related to the angular velocity $\vec{w}$. For the purpose of representing the attitude dynamics, let $\mathbf{w}$ be the quaternion for $\vec{\omega}$, then the time derivative of quaternion, called its rate, is given by

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \circ \mathbf{w} \tag{5}$$

Note that in the above equation the angular velocity is expressed in the local body frame of reference.

III. Dynamic Modelling for Rigid Systems

In this section we state the dynamical model for a satellite. The dynamics of a fixed-pitch quadrotor is also discussed.

A. Satellite Attitude Dynamics

The actuators present in a spacecraft can generate torque in the body frame $\mathcal{B}$ about all the three axis by the application of a pair of forces opposite to each other but equal in magnitude and acting in a direction normal to the line joining the actuators.[8] If the angular velocity is denoted by $\vec{\omega}$ and the quaternion by $\mathbf{q}$, the attitude dynamics of satellite is governed by the equations:

$$\mathbf{J}\dot{\vec{\omega}} = -\vec{\omega} \times \mathbf{J}\omega + \vec{M} \tag{6}$$

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \circ \mathbf{w} \tag{7}$$

where $\mathbf{J}$ is the moment of inertia tensor of the body and $\vec{M}$ is the moment generated by the actuators present on the spacecraft. In our work, we will not be dealing with position control of spacecrafts and hence the related dynamics are not mentioned here.

B. Fixed-Pitch Quadrotor Dynamics

The Newton-Euler equations of motion for a rigid body can be used to describe the motion of Quadrotor in a three dimensional space. Let $\vec{r}$ denote the position of center of mass of the quadrotor in the inertial or world frame. The axis systems chosen are as shown in Fig 1 with the body and inertial world frame Z axis directed downwards. $\vec{F}^b$, which is the force produced by rotors in the body frame, is represented by $\vec{F}^i$ in the inertial frame. Note that $\vec{F}^b$ acts along the body z axis, i.e., $\vec{F}^b = [0\ 0\ f]^T$ where f is the total thrust produced by the rotors. The gravitational force is taken to act along the positive z direction of inertial frame through the center of mass of the symmetrical quadrotor. The dynamics are then given by:[12]

$$\ddot{\vec{r}} = \frac{1}{m}\vec{F}^i - \vec{g} \tag{8}$$

$$\mathbf{J}\dot{\vec{\omega}} = \vec{M}^b - \vec{\omega} \times \mathbf{J}\vec{\omega} \tag{9}$$

where $\vec{\omega}$ is the angular velocity in the body frame and $\mathbf{J}$ is the moment of inertia tensor in the body axis system. Given a rotation matrix, $\mathbf{R}_{\mathcal{B}}^{\mathcal{E}}$, which operates from Body frame to inertial Earth frame, the vector in inertial frame can be written in terms of that in body frame as $\vec{F}^i = R_{\mathcal{B}}^{\mathcal{E}}\vec{F}^b$.

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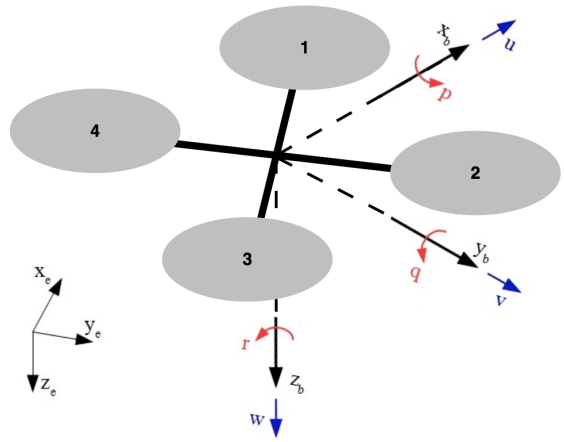

Figure 1: Inertial and body frame of reference

1. *Equations of Motion*

The translation and attitude kinematics and dynamics from Eq.9 can be recast using the quaternion method of attitude representation as

$$\ddot{\mathbf{r}} = \frac{1}{m}\mathbf{q} \circ \mathbf{F}^b \circ \mathbf{q}^* - \mathbf{g} \tag{10}$$

$$\mathbf{J}\dot{\vec{\omega}} = \vec{M}^b - \vec{\omega} \times \mathbf{J}\vec{\omega} \tag{11}$$

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \circ \mathbf{w} \tag{12}$$

2. *Rotor Dynamics*

The thrust and torque of each rotor can be obtained from Blade element theory and the momentum theory as a function of the thrust coefficient. Given the density of air (ρ), the rotor disk area (A), blade rotation angular speed (Ω), and hence the rotor tip speed $V_{tip} = \Omega R$, the non-dimensional coefficients are defined in the usual way as $C_T = T/(\rho A V_{tip}^2/2)$ and $C_Q = Q/(\rho A R V_{tip}^2/2)$. Following the same approach as in work,[13–16] the non-dimensional thrust coefficient C_{T_i}, and torque coefficient C_{Q_i}, for the ith rotor are given by

$$C_{T_i} = \frac{1}{2}\sigma C_{l\alpha}\left(\frac{\theta_{0_i}}{3} - \frac{\lambda_i}{2}\right) \tag{13}$$

$$C_{Q_i} = \frac{1}{2}\sigma\left(\frac{\lambda_i C_{l\alpha}\theta_{0_i}}{3} - \frac{\lambda_i^2 C_{l\alpha}}{2} + \frac{C_{d_{0_i}}}{4}\right) \tag{14}$$

where $C_{l\alpha}$ is the lift curve slope, θ_{0_i} is blade collective pitch angle of the ith rotor, and $C_{d_{0_i}}$ is the zero lift drag coefficient of the airfoil of the ith rotor. σ is defined in terms of number of blades N_b, chord length of the rotor c, and the rotor blade radius R as $\sigma = \frac{N_b c}{\pi R}$. The only unknown in the above equations is the inflow factor, λ_i, which can be approximated from the momentum theory for hovering flight condition as

$$\lambda_i = \sqrt{\frac{C_{T_i}}{2}} \tag{15}$$

The above substitution gives us the collective pitch input to the rotor and the torque coefficient for each rotor as function of C_T

$$\theta_{0_i} = \frac{6C_{T_i}}{\sigma C_{l\alpha}} + \frac{3}{2}\sqrt{\frac{C_{T_i}}{2}} \tag{16}$$

$$C_{Q_i} = \frac{1}{2}\sigma\left(\frac{\sqrt{2}C_{T_i}^{3/2}}{\sigma} + \frac{C_{d_{0_i}}}{4}\right) \tag{17}$$

The thrust and the torque for the ith rotor are given by

$$T_i = KC_{T_i} \tag{18}$$

$$Q_i = KRC_{T_i} \tag{19}$$

where $K = \rho\pi R^2 V_{tip}^2$ which is a constant if the RPM is maintained constant.

$$T = (T_1 + T_2 + T_3 + T_4) \tag{20}$$

$$= K(C_{T_1} + C_{T_2} + C_{T_3} + C_{T_4}) \tag{21}$$

Now given the respective moment arms, d, of rotors, the rolling and pitching moments can also be obtained. Due to the relative sense of rotations, the rotors 1 and 3 generate torque along the positive body z axis while rotors 2 and 4 generate torque about negative body z axis. The final expression for total moments acting on the quadrotor are as shown in equations below.

$$l = d \times K(C_{T_1} - C_{T_2} - C_{T_3} + C_{T_4}) \tag{22}$$

$$m = Kd(C_{T_1} + C_{T_2} - C_{T_3} - C_{T_4}) \tag{23}$$

$$n = \frac{KR}{\sqrt{2}}(|C_{T_1}|^{3/2} - |C_{T_2}|^{3/2} + |C_{T_3}|^{3/2} - |C_{T_4}|^{3/2}) \tag{24}$$

IV. Attitude Control

We impose second order error stabilizing dynamics for achieving the desired attitude. The quaternion error is defined as:

$$\mathbf{q}_e = \mathbf{q}_d^* \circ \mathbf{q} \tag{25}$$

where $\mathbf{q}^*$ denotes the conjugate or inverse of the unit quaternion. It can be interpreted as the rotation corresponding to transformation from local to desired frame of reference. The equilibrium point for the error quaternion is $\mathbf{q}_e = [1\ 0\ 0\ 0]^T$ which corresponds to perfect overlap between desired and actual orientations. The second order dynamics imposed on each component of $\mathbf{q}_e$ separately can be written together in vector form as:

$$\ddot{\mathbf{q}}_e + 2\zeta\omega_n\dot{\mathbf{q}}_e + \omega_n^2\left(q_e - \begin{bmatrix}1\\0\\0\\0\end{bmatrix}\right) = 0 \tag{26}$$

where ζ and ω_n are diagonal matrices given by:

$$\zeta = \begin{bmatrix}\zeta_1 & 0 & 0 & \\ 0 & \zeta_2 & 0 & 0\\ 0 & 0 & \zeta_3 & 0\\ 0 & 0 & 0 & \zeta_4\end{bmatrix},\ \omega = \begin{bmatrix}\omega_1 & 0 & 0 & \\ 0 & \omega_2 & 0 & 0\\ 0 & 0 & \omega_3 & 0\\ 0 & 0 & 0 & \omega_4\end{bmatrix} \tag{27}$$

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1. Quaternion Error Dynamics

From $\mathbf{q} = \mathbf{q}_d \circ \mathbf{q}_e$, we get

$$\dot{\mathbf{q}}_e = \mathbf{q}_d^* \circ (\dot{\mathbf{q}} - \dot{\mathbf{q}}_d \circ \mathbf{q}_e) \tag{28}$$

Now Using Eq.(5) for both $\mathbf{q}$ and $\mathbf{q}_d$, we get

$$\begin{aligned}
\dot{\mathbf{q}}_e &= \mathbf{q}_d^* \circ (\frac{1}{2}\mathbf{q} \circ \mathbf{w} - \frac{1}{2}\mathbf{q}_d \circ \mathbf{w}_d^D \circ \mathbf{q}_e) && (29)\\
&= \frac{1}{2}(\mathbf{q}_d^* \circ \mathbf{q} \circ \mathbf{w}_B - \mathbf{w}_d^D \circ \mathbf{q}_e) && (30)\\
&= \frac{1}{2}(\mathbf{q}_e \circ \mathbf{w} - \mathbf{w}_d^D \circ \mathbf{q}_e) && (31)\\
&= \frac{1}{2}\mathbf{q}_e \circ (\mathbf{w} - \mathbf{q}_e^* \circ \mathbf{w}_d^D \circ \mathbf{q}_e) && (32)
\end{aligned}$$

where $\mathbf{w}_d$ is the desired angular velocity quaternion and D stands for desired quaternion frame. Noting that B denotes body frame and that by definition $\mathbf{q}_e$ operates on a vector in body(actual) quaternion frame and maps it to desired attitude frame, the above can be simplified in terms of angular velocity error $\tilde{w}$ as

$$\begin{aligned}
\mathbf{w}_d^D = q_e \circ \mathbf{w}_d^B \circ \bar{q}_e \implies \dot{\mathbf{q}}_e &= \frac{1}{2}\mathbf{q}_e \circ (\mathbf{w} - \mathbf{w}_d^B) && (33)\\
&= \frac{1}{2}\mathbf{q}_e \circ \tilde{\mathbf{w}} && (34)
\end{aligned}$$

The second derivative of quaternion would naturally involve angular acceleration terms. Writing Eq.(34) as $\tilde{\mathbf{w}} = 2\mathbf{q}_e^* \circ \dot{\mathbf{q}}_e$, we obtain

$$\begin{aligned}
\dot{\tilde{\mathbf{w}}} &= 2\dot{\mathbf{q}}_e^* \circ \dot{\mathbf{q}}_e + 2\mathbf{q}_e^* \circ \ddot{\mathbf{q}}_e && (35)\\
&= 2(||\mathbf{q}_e||^2, 0) + 2\mathbf{q}_e^* \circ \ddot{\mathbf{q}}_e && (36)
\end{aligned}$$

The vector part of the error quaternion, $\dot{\vec{\tilde{\omega}}}$ in $\dot{\tilde{\mathbf{w}}} = \begin{bmatrix} 0 & \vec{\tilde{\omega}}^T \end{bmatrix}^T$ is therefore

$$\begin{aligned}
\dot{\vec{\tilde{\omega}}} &= 2\begin{bmatrix} -q_1 & q_0 & q_3 & -q_2 \\ -q_2 & -q_3 & q_0 & q_1 \\ -q_3 & q_2 & -q_1 & q_0 \end{bmatrix}_e \ddot{\mathbf{q}}_e && (37)\\
&= 2G\ddot{\mathbf{q}}_e && (38)
\end{aligned}$$

where the matrix G is defined as above for convenience.

2. Controller

We can get the desired local angular acceleration from the error dynamics using Eqs.(38) and (26)

$$\begin{aligned}
\dot{\vec{\omega}}_D &= \dot{\vec{\omega}} + \dot{\vec{\omega}}_d^B && (39)\\
&= 2G(-2\zeta_i\omega_{n_i}\dot{\mathbf{q}}_e - \omega_{n_i}^2\left(\mathbf{q}_e - [1\ 0\ 0\ 0]^T\right)) + \dot{\vec{\omega}}_d^B && (40)
\end{aligned}$$

where $\vec{\omega}_D$ is the desired angular velocity rate which is pass to the control allocation as the angular acceleration to be achieved. The required moments and thrust can be obtained from the analysis till now which then can be converted to required rotor speeds directly using the transformation matrix.

Finally note that in Eq.(40), the derivative of the desired angular velocity in body frame $\dot{\vec{w}}_d^B$ can be obtained using $\vec{\omega}_d^B = R(\mathbf{q}_e)^T\vec{\omega}_d^D$ and hence

$$\dot{\vec{\omega}}_d^B = \dot{R}^T\vec{\omega}_d^D + R^T\dot{\vec{\omega}}_d^D \tag{41}$$

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V. Trajectory Tracking

This section develops a Nonlinear Dynamic Inversion(NDI)[17,18] based controller for the quadrotor. We propose a controller with one outer and one inner layer loop for position and attitude control.

A. Outer Loop

Since the quadrotors are differentially flat,[3] the four flat outputs, which are three spatial positions and the yaw angle, together completely determine the required four thrust command for rotors. Let $\vec{r}_d$ be the desired quadrotor position in the inertial frame and $\vec{e} = \vec{r} - \vec{r}_d$ be the error in position. In accordance with the NDI methodology, we choose second order error stabilizing dynamics for translation as

$$\ddot{\vec{e}} + 2\zeta_p \omega_{n_p} \dot{\vec{e}} + \omega_{n_p}^2 \vec{e} = 0 \tag{42}$$

where ζ and ω_n are design matrices. Using the fact that the only forces that act on quadrotor are net thrust force and gravity, the above equation can be used to compute the desired attitude and net thrust T.

$$\ddot{\vec{r}} = \ddot{\vec{r}}_d + 2\zeta\omega_n(\dot{\vec{r}}_d - \dot{\vec{r}}) + \omega_n^2(\vec{r}_d - \vec{r}) \tag{43}$$

$$T_d = |\ddot{\vec{r}} - \vec{g}| \tag{44}$$

where T_d is the desired thrust, $\vec{g} = [0\ 0\ g]$ and we have used the fact that only force acting on quadrotor besides the gravitational force is the rotor thrust. The desired quaternion now is the one that relates the desired thrust in body and inertial frame. Following on similar lines to work done by Cutler and How,[19] we get from Eq.(12)

$$\mathbf{F}^i = m(\ddot{\mathbf{r}} - \mathbf{g}) = \mathbf{q}_{d'} \circ \mathbf{F}^b \circ \mathbf{q}_{d'}^* \tag{45}$$

where $q_{d'}$ is the desired quaternion that orients the thrust vector with the desired inertial frame force. It is equal to the minimum angle rotation between the two unit vectors $\hat{F}^i$ and $\hat{F}^b$ given by:[20]

$$q_{d'} = \frac{1}{\sqrt{2(1 + \hat{F}^b \cdot \hat{F}^i)}} \begin{bmatrix} 1 + \hat{F}^b \cdot \hat{F}^i \\ \hat{F}^b \times \hat{F}^i \end{bmatrix} \tag{46}$$

where $||\vec{F}^b|| = ||\vec{F}^i|| = T$. Note that $q_{d'}$ takes into account only the roll and pitch angles since any arbitrary yaw motion would still align the vectors in body and inertial frame. Also, $\hat{F}^b = [0\ 0\ \pm 1]$ depending on the direction of thrust vector. In case of fixed-pitch quadrotor we have one direction and hence $\hat{F}^b = [0\ 0\ -1]$. Also, note that due to dual covering of attitude space, that is, since $\mathbf{q}$ and $-\mathbf{q}$ represent the same quaternion, there can be a discontinuity in the control law. Therefore we perform a check so that current time step quaternion is closest to the previous time step quaternion, that is, $\mathbf{q}_{d_{t-1}}^T \mathbf{q}_{d_t} \geq 0$.

Finally since the desired yaw angle is given as input from trajectory generation algorithm, the desired quaternion can be obtained by complementing $q_{d'}$ with ψ_d in following way:[19]

$$q_d = q_{d'} \circ [\cos(\psi_d/2)\ 0\ 0\ \sin(\psi_d/2)] \tag{47}$$

B. Inner Loop

The output of the outer loop is passed to the attitude controller described in Section IV. The required moment can now be obtained from Eq.(12). Note that the desired angular velocity rate required in the attitude controller can be obtained numerically from desired attitude history. We follow the same procedure as done by Cutler[19] and hence omit it here.

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VI. Simulation Results

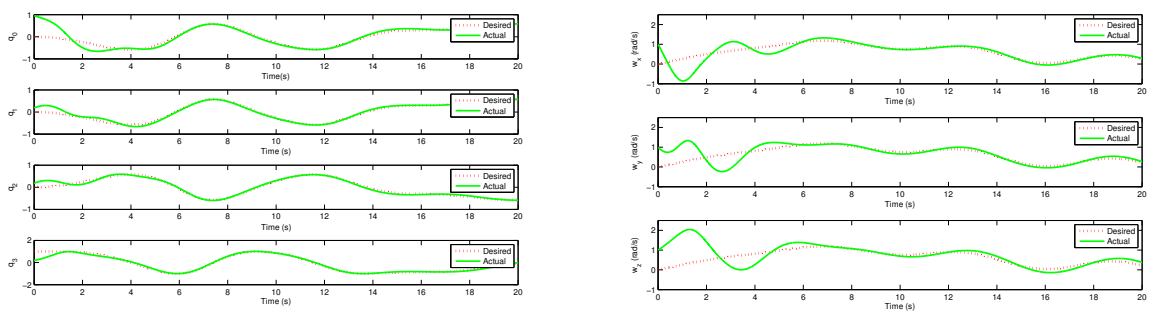
In this section, we present numerical results to show the effectiveness of the proposed algorithm. The performance is demonstrated by two examples. First we show results for attitude tracking by taking example of a sample satellite attitude history. Next we show results for trajectory tracking of a fixed-pitch fixed pitch quadrotor.

A. Stabilization and Attitude Tracking

First the capability of the attitude controller is demonstrated by tracking a sample attitude history. The following angular velocity is taken from work by Costic et al.[8]

$$w_d(t) = \begin{pmatrix} 0.3\cos(t)(1-e^{-0.01t^2}) + [0.08\pi + 0.006\sin(t)]te^{-0.01t^2} \\ 0.3\cos(t)(1-e^{-0.01t^2}) + [0.08\pi + 0.006\sin(t)]te^{-0.01t^2} \\ 0.3\cos(t)(1-e^{-0.01t^2}) + [0.08\pi + 0.006\sin(t)]te^{-0.01t^2} \end{pmatrix} \tag{48}$$

The initial quaternion is chosen as $\mathbf{q}_{t-9} = [0.94846\ 0.1826\ 0.1826\ 0.1826]^T$ and the desired quaternion history is obtained from above angular velocity and taking $\mathbf{q}_{d_{t=0}}$ as $[1\ 0\ 0\ 0]^T$. The values of gain matrices is chosen such that $\zeta_i = 0.3$ and $\omega_n = diag(0.5, 2, 21.5)$. Fig.2a shows the desired and actual quaternion trajectory and Fig.2b shows the desired and actual angular velocity. It can be seen that both the desired attitude and angular velocity is achieved within 6 seconds.



(a) Quaternion tracking using attitude control

(b) Angular velocity tracking using attitude control

Figure 2

B. Fixed-Pitch Quadrotor

We demonstrate trajectory tracking for a fixed-pitch quadrotor using spiral trajectory shown below:

$$X_d = \begin{pmatrix} \frac{\sin(\pi t/2)}{2} \\ \frac{\sin(\pi t/2)}{2} \\ t \end{pmatrix} \tag{49}$$

the gain matrices are chosen as $\zeta_i = 0.08$, $\omega_n = diag(2, 8, 8, 6)$ for the attitude loop and $\zeta_{p_i} = 0.1$ and $\omega_{p_n} = 1$. The initial state is $\vec{r}_{t=0} = (0\ 0\ 0)^T$, $\dot{\vec{r}} = (0\ 0\ 0)^T$, $\mathbf{q} = (1\ 0\ 0\ 0)^T$ and $\vec{\omega} = (0\ 0\ 0)^T$. The time step chosen for simulation is 0.001s and the position loop is run at 100 Hz while the attitude loop is run at 1000Hz. Fig.3 shows the desired and actual position of the quadrotor as a function of time. Fig.5a gives a 3 dimensional visualization of the same path. Fig.4a shows the Euler angle variation just for better understanding as they are more intuitive and gives a better idea of orientation at any point of time. Fig.4b shows the quaternion variations, which are the actual mode or representing attitude in our work. Finally, Fig.5b shows the thrust variation with time. From above plots, it is clear that the proposed controller is able to track the desired trajectory effectively. The rate of convergence is a matter of tuning and higher gains can be chosen for faster tracking.

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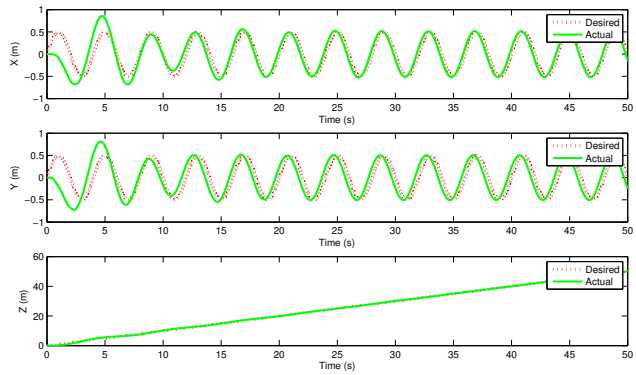
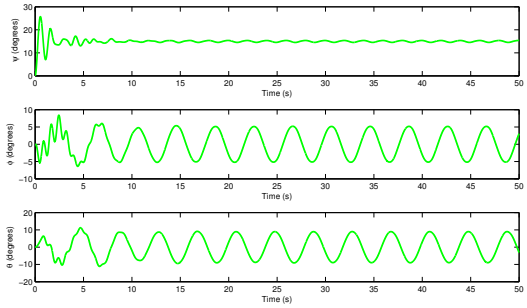
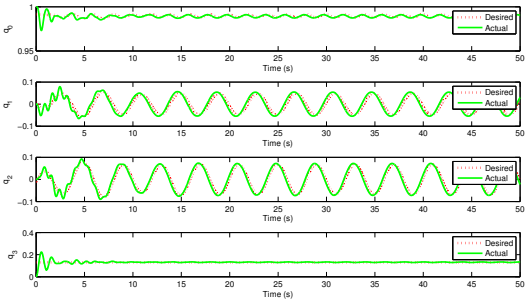


Figure 3: Quadrotor position control

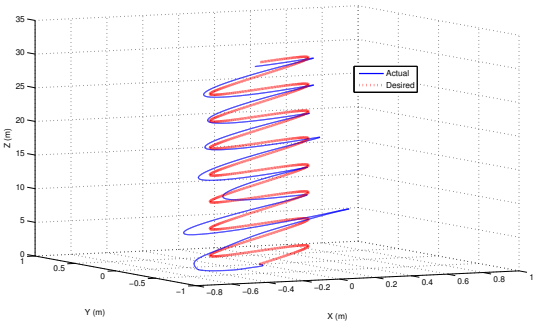


(a) Euler angles variation

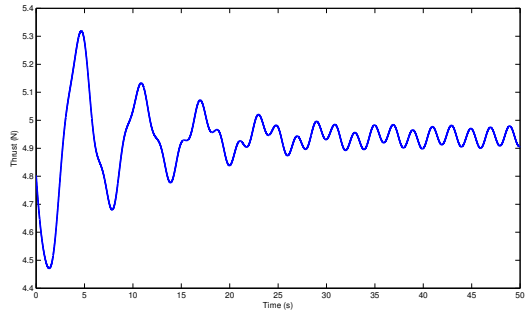


(b) Quaternion tracking

Figure 4

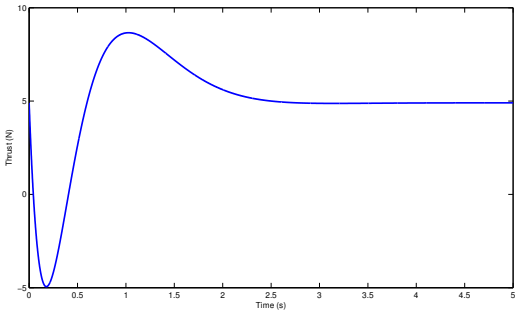


(a) 3D Visualization of Quadrotor spiral path during trajectory tracking control

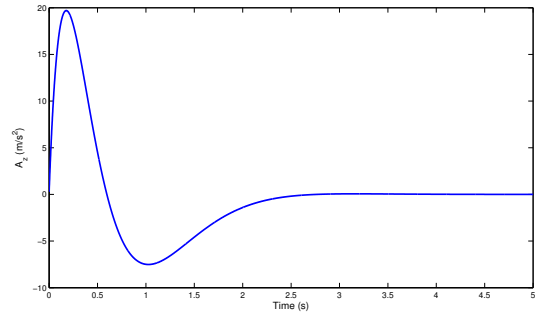


(b) Quadrotor thrust as function of time

Figure 5



(a) Quadrotor thrust variation with time



(b) Quadrotor achieved vertical acceleration

Figure 6

VII. Experimental Validation

In this section we discuss the experimental results using the proposed controller. Details of the experimental setup followed by results of attitude tracking are presented.

A. Hardware

The proposed control algorithm is implemented on a commercially available fixed pitch H-King Color 250 Class quadrotor shown in fig 7.

We use the PX4[21] opensource codebase running on Pixhawk autopilot hardware. Pixhawk uses ST Micro L3GD20H 16 bit gyroscope,ST Micro LSM303D 14 bit accelerometer / magnetometer and Invensense MPU 6000 3-axis accelerometer/gyroscope as sensors. These sensors are used in a non-linear filter to estimate the attitude of the quadrotor.The codebase uses a PID controller for attitude which is totally replaced with our quarternion based controller. The attitude loop is run as a priority thread at 250Hz with the actuation commands being given to the speed controllers at 50Hz. The quadrotor is flown in manual mode in which the pilot generates the (T ,θ , ϕ , ψ) as control inputs. Using equation 51, the control input was converted to a quaternion representation (q_d).A 915MHz telemetry module is used to remotely communicate with the ground station.An FrSky X9D transmitter is used as the radio controller with FrSky X8R receiver placed onboard the vehicle and paired with the transmitter.The overall takeoff weight for the quadcopter is 600 gms. Moment of Inertia along I_{xx}= 2.2686E-3 kgm^2, I_{yy}=2.24557E-3 kgm^2 , I_{zz}=3.19819E-3 kgm^2. The cross moment of inertia are assumed to be zero as it is a nearly symmetrical structure.

$$q = q_\phi \circ q_\theta \circ q_\psi \tag{50}$$

$$= \begin{bmatrix} \cos(\phi/2)\cos(\theta/2)\cos(\psi/2) - \sin(\phi/2)\sin(\theta/2)\sin(\psi/2) \\ \cos(\phi/2)\sin(\theta/2)\sin(\psi/2) + \sin(\phi/2)\cos(\theta/2)\cos(\psi/2) \\ \cos(\phi/2)\cos(\psi/2)\sin(\theta/2) - \sin(\phi/2)\cos(\theta/2)\sin(\psi/2) \\ \cos(\phi/2)\cos(\theta/2)\sin(\psi/2) + \cos(\psi/2)\sin(\theta/2)\sin(\phi/2) \end{bmatrix} \tag{51}$$

B. Experiment

The experiments are performed outdoor on the IIT Kanpur airstrip with wind speed of around 2m/s. The quadrotor is flown manually and the values of ζ and ω_n are tuned so as to get a sufficient attitude tracking performance. The system is able to track attitude commands with no position constraints. This is verified with a series of test flights with manual radio controller inputs. One such flight data is shown in Fig.8.The gain values are tabulated in Table 1. Small roll oscillations are given as input to showcase tracking.[Video Link =https://youtu.be/4bJHLSYMYxw]

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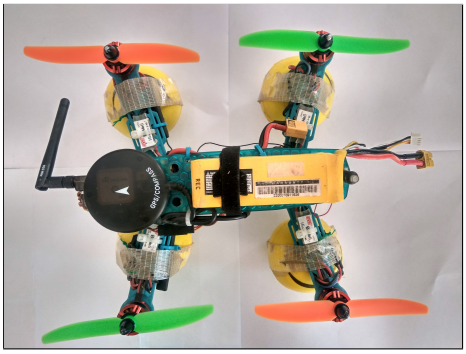

Figure 7: Fixed pitch quadrotor used for experiments

	Parameter	Value
ζ	ζ_{q0}	0.85
	ζ_{q1}	0.86
	ζ_{q2}	0.86
	ζ_{q3}	0.86
ω	ω_{q0}	10.65
	ω_{q1}	10.65
	ω_{q2}	11
	ω_{q3}	13

Table 1: Experimentally obtained tuning parameters

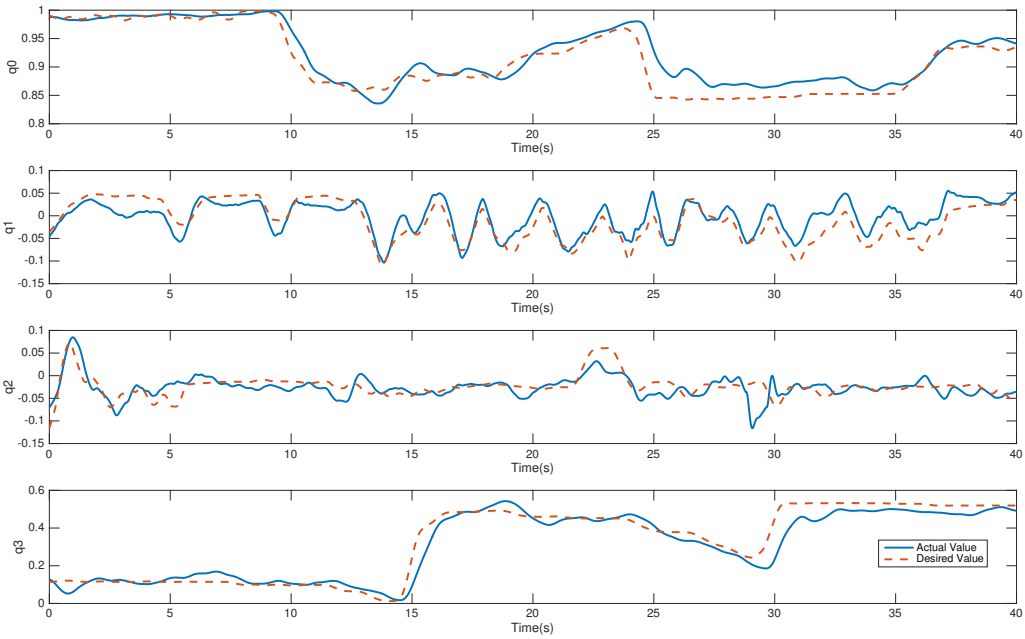

Figure 8: Quaternion tracking in Manual Mode

VIII. Conclusions

In this paper, we have presented theory, implementation and experimental results for a novel second order stable error dynamics controller using a fully quaternion based approach. The suitability of the generic controller for highly non-linear systems like quadrotors and satellites is examined by designing controllers that efficiently track the desired trajectories. It avoids the problem of singularity often encountered in designing controllers for satellites with the added benefit of a second order system generating smoother control inputs. A framework for designing attitude controllers for rigid bodies and position controller for quadrotors is discussed and numerical simulation results are presented.

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